

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1374

Commonwealth Chesapeake 138 kV
March 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1374. The Interconnection Customer proposes a 79.6 MW solar photovoltaic facility interconnecting at the Commonwealth Chesapeake 138 kV substation in the Virginia area. The request was received on January 03, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 332 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$417,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ST BONIFACIUS-W -STATION SUBSTATION 138kV	VIENNA OPERATIONS-SOUTHTOWN 345kV line	237.9	417	53.6
RANDALL COUNTY-CHEMOLITE 138kV line	VIENNA OPERATIONS SUBSTATION-EASTON 2 345kV	101.8	230	15.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ST BONIFACIUS-W -STATION SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$9,035,000	3% (\$310,000)
CHEMOLITE-RANDALL COUNTY SUBSTATION 138kV	Reconductor line	\$3,040,000	4% (\$107,000)
	Total allocated to project		\$417,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.